Pregnancy Risk Prediction

Model Comparison Report

Project Scope

Compare 4 classifiers for
pregnancy risk levels:
LOW, MEDIUM, HIGH

Dataset

3600 synthetic rows
Balanced classes
Noise + overlap + medical interactions

Compared Models

- Random Forest
- Logistic Regression
- SVM
- XGBoost

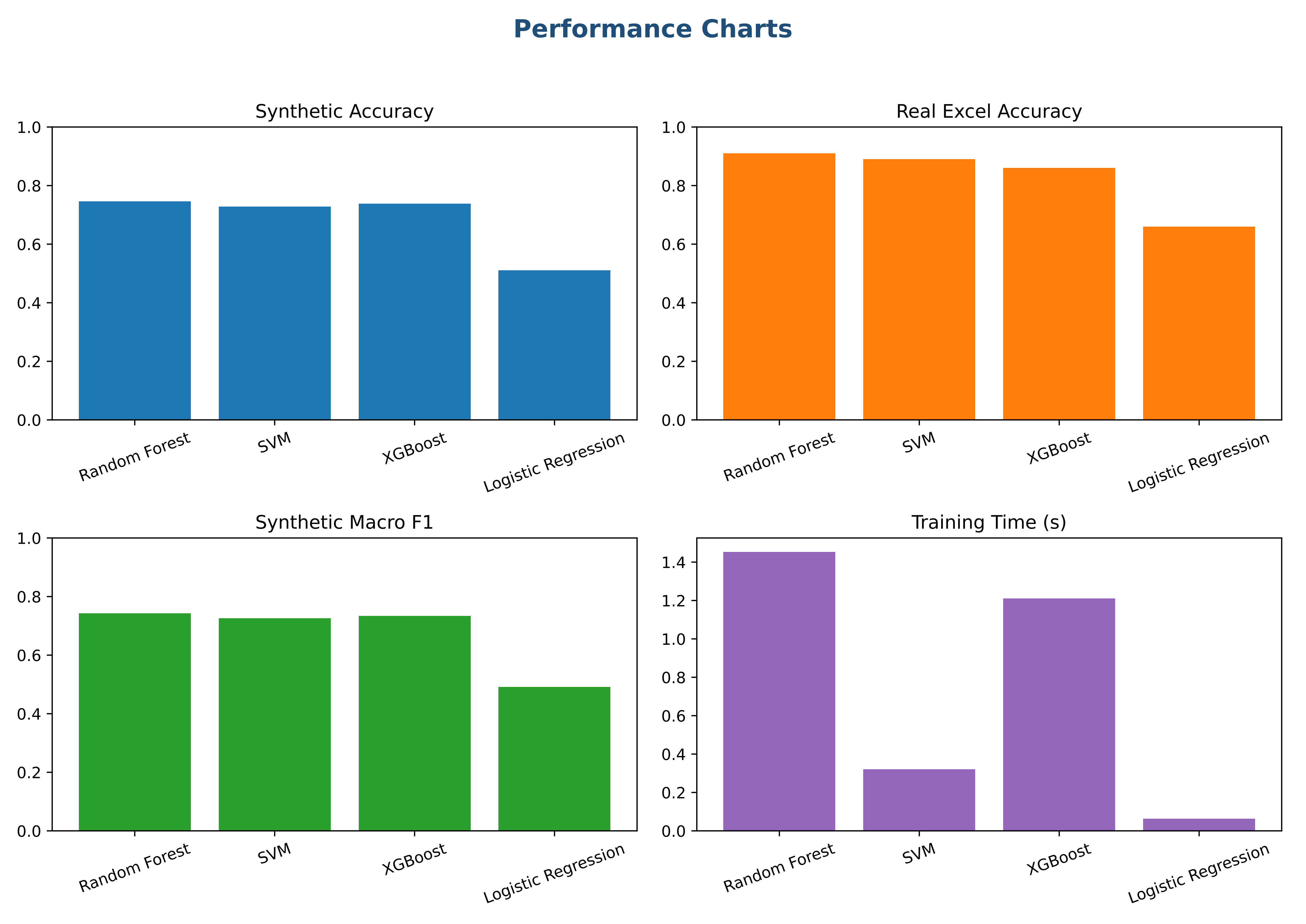
Evaluation

- Synthetic split: 70/30 (stratified)
- Real-patient Excel validation
- Accuracy, F1 per class, training time

Side-by-Side Model Comparison

Model	Train Time(s)	Syn Acc	Real Acc	F1 HIGH	F1 LOW	F1 MEDIUM
Random Forest	1.453	0.746	0.910	0.846	0.779	0.601
SVM	0.321	0.728	0.890	0.828	0.766	0.582
XGBoost	1.210	0.738	0.860	0.848	0.769	0.585
Logistic Regression	0.064	0.510	0.660	0.691	0.504	0.277

Best Real Validation Accuracy: Random Forest


Performance Charts
Synthetic Accuracy
Real Excel Accuracy
Synthetic Macro F1
Training Time (s)
Random Forest
SVM
XGBoost
Logistic Regression
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4

Model-by-Model Summary

Random Forest
Train Time: 1.453s | Synthetic Acc: 0.746 | Real Acc: 0.910
Synthetic F1 -> HIGH: 0.846, LOW: 0.779, MEDIUM: 0.601
Real F1 -> HIGH: 0.885, LOW: 0.982, MEDIUM: 0.667

SVM
Train Time: 0.321s | Synthetic Acc: 0.728 | Real Acc: 0.890
Synthetic F1 -> HIGH: 0.828, LOW: 0.766, MEDIUM: 0.582
Real F1 -> HIGH: 0.828, LOW: 0.991, MEDIUM: 0.645

XGBoost
Train Time: 1.210s | Synthetic Acc: 0.738 | Real Acc: 0.860
Synthetic F1 -> HIGH: 0.848, LOW: 0.769, MEDIUM: 0.585
Real F1 -> HIGH: 0.848, LOW: 0.946, MEDIUM: 0.552

Logistic Regression
Train Time: 0.064s | Synthetic Acc: 0.510 | Real Acc: 0.660
Synthetic F1 -> HIGH: 0.691, LOW: 0.504, MEDIUM: 0.277
Real F1 -> HIGH: 0.767, LOW: 0.771, MEDIUM: 0.273

Final Recommendation

Conclusion

Best model on real-patient Excel validation: Random Forest.

Random Forest remains a strong deployment choice in this PFE context because it provides a practical balance:
- Strong and stable performance across classes
- Simple workflow with little parameter tuning
- Robustness to noisy and overlapping medical patterns
- Reliable behavior on small and medium datasets

This makes Random Forest a safe baseline model for student-level clinical risk prototyping.